Properties of the thin radiating surface layer of a white dwarf

Riccardo Fantoni*
Università di Trieste, Dipartimento di Fisica, strada Costiera 11, 34151 Grignano (Trieste), Italy
(Dated: August 15, 2026)

Keywords: White dwarf;Jellium;Path Integral;Monte Carlo;General Relativity;Thin surface layer

CONTENTS

I. INTRODUCTION

White dwarfs [1] are the dense remnants of stars, composed mainly of carbon and oxygen nuclei, which contain electrons, protons and neutrons and are supported against collapse induced by gravity by the electron degeneracy pressure due to Pauli exclusion principle. They remain white dwarfs till gravity overcomes electron degeneracy pressure, crushing electrons into protons to form neutrons through β^+ decay.

Therefore the simplest model for the interior of a white dwarf is that of *Jellium*. This is a system of pointwise electrons immersed in a uniform neutralizing background. The density of the electron gas is so high that the Fermi temperature is much higher than the temperature of the star and the electron gas can be considered degenerate. Chandrasekar [2–5] model of a white dwarf equilibrium structure didn't take into account the effects of general relativity. First Tolman [6], and later Oppenheimer and Volkoff [7] proposed a way to treat the hydrodynamic equations for the equilibrium and stability of a star within the framework of general relativity.

The luminosity of a white dwarf is due to a thin radiating layer on the star surface (see chapter 4 of the book [1]). The mechanism of a white dwarf cooling is the primary bridge between our theoretical description of the star and observational data which rely on essentially two aspects: luminosity observations and time scale of the observed cooling.

Purpose of this work is to determine the properties of the electron plasma in this thin surface layer using the path integral Monte Carlo (PIMC) method [8] at low non zero temperature. In order to take into account the effects of general relativity we will use the $3+1$ foliation of Arnowitt, Deser, and Misner (ADM) [9] to build a density matrix from the stress-energy tensor through a path integral where on each imaginary timeslice one selects a constant time foil [10]. This amounts to treat the quantum Coulomb plasma in the classical Schwarzschild spatial metric outside a spherical shell containing a mass $\mathscr{M}$ as confined in a thin spherical layer of width $\delta \ll \mathscr{R}$ at the surface of the star of radius $\mathscr{R}$. We will then approximate $\mathscr{M}$ as the white dwarf mass.

* riccardo.fantoni@scuola.istruzione.it

The problem of the plasma on the surface of the star, neglecting the radial thickness of the thin layer, has already been treated in Refs. [11–13]. Here we will consider the properties of the 'atmosphere' of a white dwarf.

We will also assume that the electrons dispersion relation is the one of non relativistic particles $\epsilon(p) = p^2/2m$. This is reasonable since the density of the electron-proton plasma [1] in the thin surface layer is $\rho_l \approx 10^3 \mathrm{g/cm}^3$ (see section 4.1 of the book [1]) much lower than the central densities. Therefore $p_F^2/m^2c^2 \approx 0.01$ where m is the electron mass and $p_F = h(3n/4\pi)^{1/3}$ is the polarized electrons Fermi momentum.

II. ADM FOLIATION AND SCHWARZSCHILD METRIC

By Birkhoff theorem the vacuum solution of Einstein field equations outside a mass $\mathscr{M}$ is the one of Schwarzschild. Within the ADM foliation

$$||g_{\alpha\beta}|| = \begin{pmatrix} -(N^2 - N^i N_i) & N_i \\ N_i & g_{ij} \end{pmatrix}, \tag{2.1}$$

the *shift* $N_i = 0$, the *lapse* $N^2 = (1 - R_s/r)$, and $g_{rr} = (1 - R_s/r)^{-1}, g_{\theta\theta} = r^2, g_{\varphi\varphi} = r^2 \sin^2\theta$ with $R_s = 2\mathscr{M}$ the Schwarzschild radius. We will generally use a Greek index to denote one of the 4 spacetime components and a Roman index to denote one of the 3 spatial components. We will also introduce ${}^3g = \det\{g_{ij}\} > 0$.

Contracting Einstein field equations in vacuum implies a vanishing scalar curvature, the trace of Ricci tensor. Then Einstein field equations in vacuum reduce to the vanishing of the Ricci tensor. Therefore also the scalar curvature of a spatial foil, at constant time, of Schwarzschild metric must vanish. This will be used later on.

III. PIMC IN A FOIL OF SCHWARZSCHILD METRIC

IV. PIMC CONSTRAINED IN A RADIAL THIN SURFACE LAYER

V. NUMERICAL RESULTS

VI. CONCLUSIONS

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Funding

None declared.

[1] S. L. Shapiro and S. A. Teukolsky, *Black Holes, White Dwarfs, and Neutron Stars. The Physics of Compact Objects* (John Wiley & Sons Inc, New York, 1983).

[1] Assuming that the white dwarf is made up of a completely ionized plasma then we may write $\rho = \mu_e m_u n$ with n the electrons number density, m_u the atomic mass unit, and $\mu_e = A/Z$ with Z the atomic number and A the mass number, so that for example for a ^{4}He or a ^{12}C white dwarf $\mu_e = 2$ and for a ^{56}Fe white dwarf $\mu_e = 56/26 \approx 2.134$.

[2] S. Chandrasekhar and E. A. Milne, The Highly Collapsed Configurations of a Stellar Mass, Monthly Notices of the Royal Astronomical Society **91**, 456 (1931).
[3] S. Chandrasekhar, The Density of White Dwarf Stars, Phil. Mag. **11**, 592 (1931).
[4] S. Chandrasekhar, The Maximum Mass of Ideal White Dwarfs, Astrophys. J. **74**, 81 (1931).
[5] R. Fantoni, White-dwarf equation of state and structure: the effect of temperature, J. Stat. Mech. , 113101 (2017).
[6] R. C. Tolman, Static Solutions of Einstein's Field Equations for Spheres of Fluid, Phys. Rev. **55**, 364 (1939).
[7] J. R. Oppenheimer and G. M. Volkoff, On Massive Neutron Cores, Phys. Rev. **55**, 374 (1939).
[8] D. M. Ceperley, Path integrals in the theory of condensed Helium, Rev. Mod. Phys. **67**, 279 (1995).
[9] R. Arnowitt, S. Deser, and C. Misner, The Dynamics of General Relativity, In: Witten, L., Ed., Gravitation: An Introduction to Current Research, Wiley & Sons, New York, 227 (1962), arXiv: gr-qc/0405109.
[10] R. Fantoni, Formalism for the Matter Density Matrix in an ADM $3+1$ foliation of spacetime, Phys. Rev. Lett. (in preparation) (2026), `https://rfantoni.github.io/personal-page/publications/pcdms.pdf`.
[11] R. Fantoni, One-component fermion plasma on a sphere at finite temperature, Int. J. Mod. Phys. C **29**, 1850064 (2018).
[12] R. Fantoni, One-component fermion plasma on a sphere at finite temperature. The anisotropy in the paths conformations, J. Stat. Mech. , 083103 (2023).
[13] R. Fantoni, Path Integral Monte Carlo on a Sphere, Phys. Rev. B (submitted) (2026), `http://arxiv.org/abs/2603.22338`.